

1. 半径为 R 的圆环，均匀带电，总电量为 Q ，求其轴线上的电场分布（10分），电势分布（10分）。

解法一:

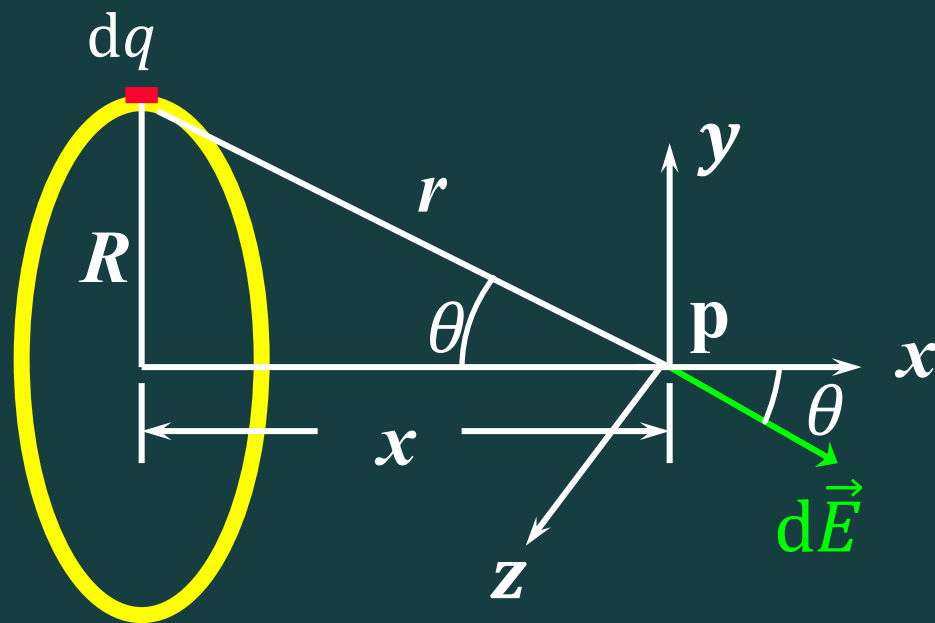
由对称性 $E_y = E_z = 0$ 2分

$$dE = \frac{1}{4\pi\epsilon_0} \frac{dq}{r^2} \quad 2分$$

$$E = E_x = \int dE \cdot \cos \theta \quad 2分$$

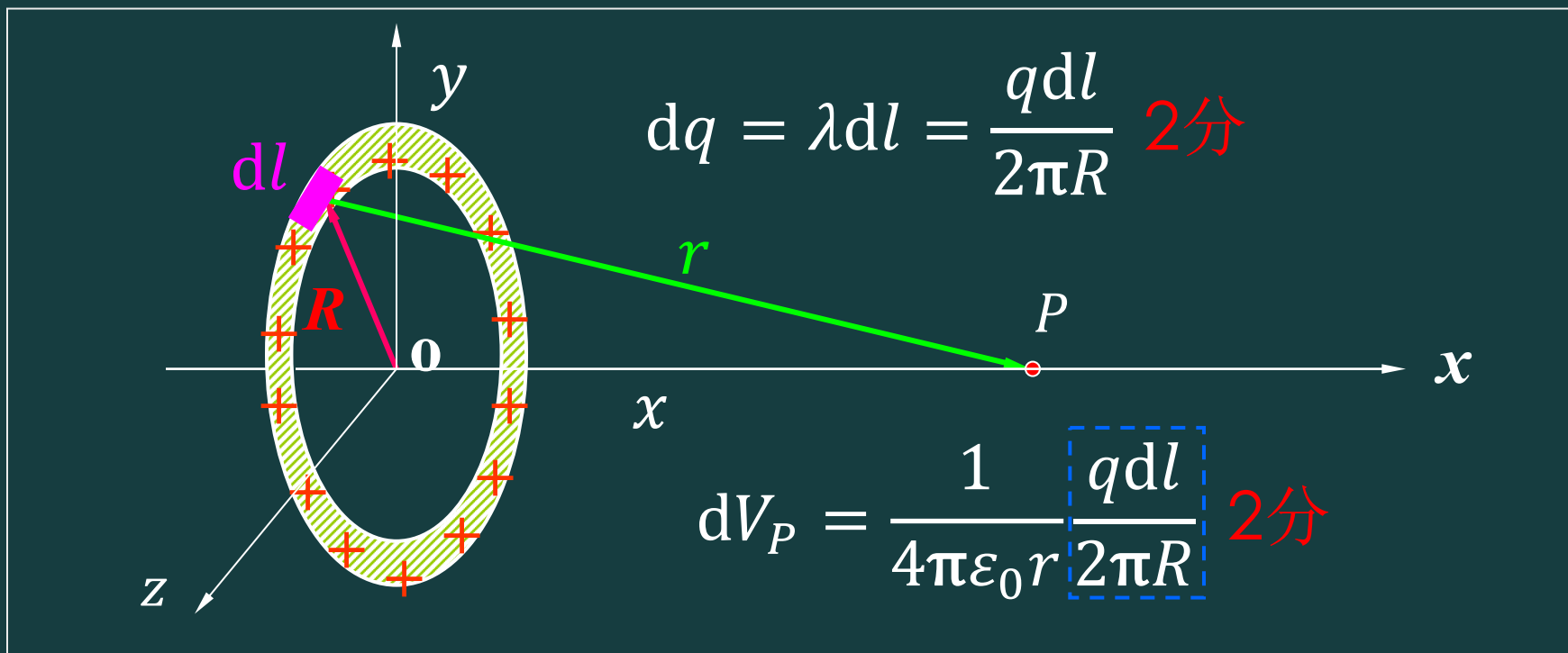
$$= \frac{\cos \theta}{4\pi\epsilon_0 r^2} \int dq = \frac{q \cos \theta}{4\pi\epsilon_0 r^2}$$

$$= \frac{q x/r}{4\pi\epsilon_0 r^2} = \frac{qx}{4\pi\epsilon_0 r^3} = \frac{qx}{4\pi\epsilon_0 (R^2 + x^2)^{3/2}} \quad 2分$$



$$\begin{aligned}
 U &= \int_x^\infty E dx \stackrel{4\text{分}}{=} \int_x^\infty \frac{qx dx}{4\pi\epsilon_0(R^2 + x^2)^{3/2}} \stackrel{4\text{分}}{=} \\
 &= \int_x^\infty \frac{q dx^2}{8\pi\epsilon_0(R^2 + x^2)^{3/2}} = \int_x^\infty \frac{q d(R^2 + x^2)}{8\pi\epsilon_0(R^2 + x^2)^{3/2}} \\
 &= -\frac{q(R^2 + x^2)^{-\frac{1}{2}}}{4\pi\epsilon_0} \bigg|_x^\infty = \frac{q(R^2 + x^2)^{-\frac{1}{2}}}{4\pi\epsilon_0} \quad 2\text{分}
 \end{aligned}$$

解法二:



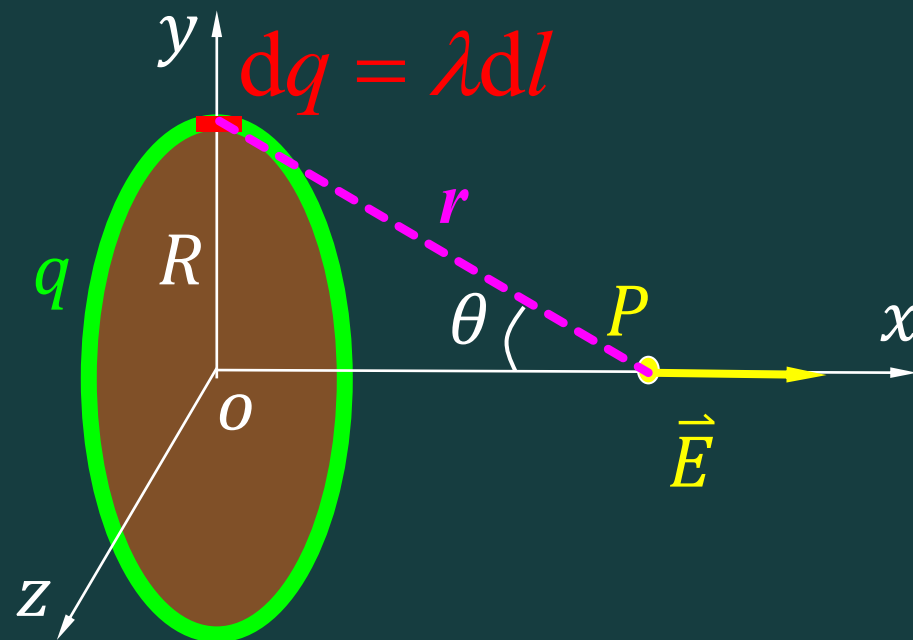
$$V_P = \frac{1}{4\pi\epsilon_0 r} \int \frac{q dl}{2\pi R} = \frac{q}{4\pi\epsilon_0 r} = \frac{q}{4\pi\epsilon_0 \sqrt{x^2 + R^2}}$$

2分 2分 2分

由对称性 $E_y = E_z = 0$ 2分

$$E = E_x = -\frac{\partial U}{\partial x} \quad 4分$$

$$= -\frac{\partial}{\partial x} \left[\frac{q}{4\pi\epsilon_0(x^2 + R^2)^{1/2}} \right] = \frac{qx}{4\pi\epsilon_0(x^2 + R^2)^{3/2}} \quad 2分$$



2. 一个平行板电容器两极板的距离 d 远小于极板的大小，中间为真空，极板面积为 S . 求电容（10分）.

(1) 设两导体板分别带 $\pm Q$ 2分

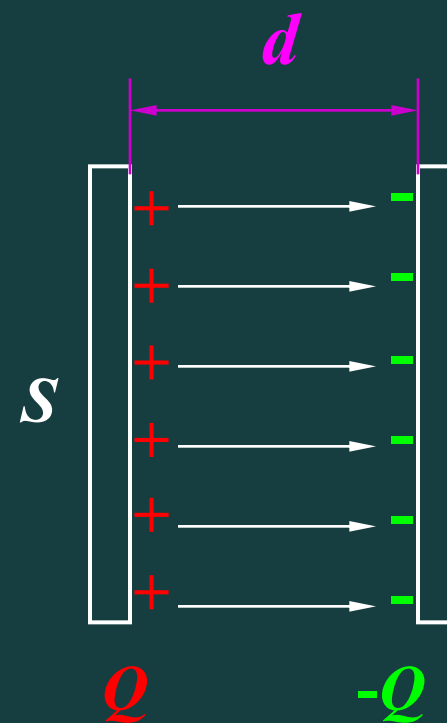
(2) 两带电平板间的电场强度

$$E = \frac{\sigma}{\epsilon_0} = \frac{Q}{\epsilon_0 S} \quad 2分$$

(3) 两带电平板间的电势差

$$U = Ed = \frac{Qd}{\epsilon_0 S} \quad 2分$$

(4) 平板电容器电容 $C = \frac{Q}{U} = \epsilon_0 \frac{S}{d}$
2分 2分



3. 一个半径为 R 的带电球，球内电荷密度正比于到球心的距离，即 $\rho = \alpha r$ ， α 为常量，求球内外的场强分布. (20分)

在球外作半径为 r 的球形高斯面，它包住的电荷量为

$$Q = \int_0^R \rho dV = \int_0^R \alpha r 4\pi r^2 dr = \alpha \pi r^4 \Big|_0^R = \alpha \pi R^4 \quad 1\text{分}$$

$$4\pi r^2 E = \frac{Q}{\epsilon_0} \quad 4\text{分}$$

$$E = \frac{\alpha \pi R^4}{4\pi r^2 \epsilon_0} \quad 2\text{分}$$

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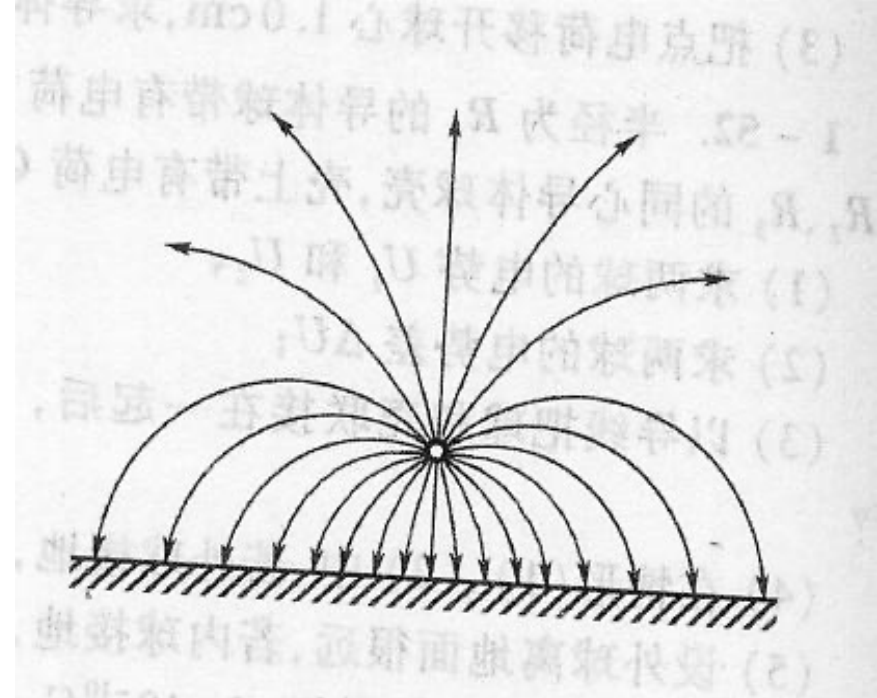
The diagram shows a positive point charge (represented by a small circle with a dot) located at a height above a horizontal line representing a grounded conducting plane. The plane is indicated by diagonal hatching below it. Numerous curved lines with arrows represent the electric field lines. These lines originate from the point charge and curve downwards, terminating at the conducting plane. The field lines are symmetric about a vertical axis passing through the charge and perpendicular to the plane. The density of the field lines is highest near the charge and decreases as they spread out towards the plane.

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4. 如图，地面可看成无穷大的导体平面，一半径为 R 均匀带电（线密度为 λ ）无限长直导线平行地面放置，与地面距离为 H ($H \gg R$)，求空间中的电势分布（10分），求地表面上的电荷分布（10分）。



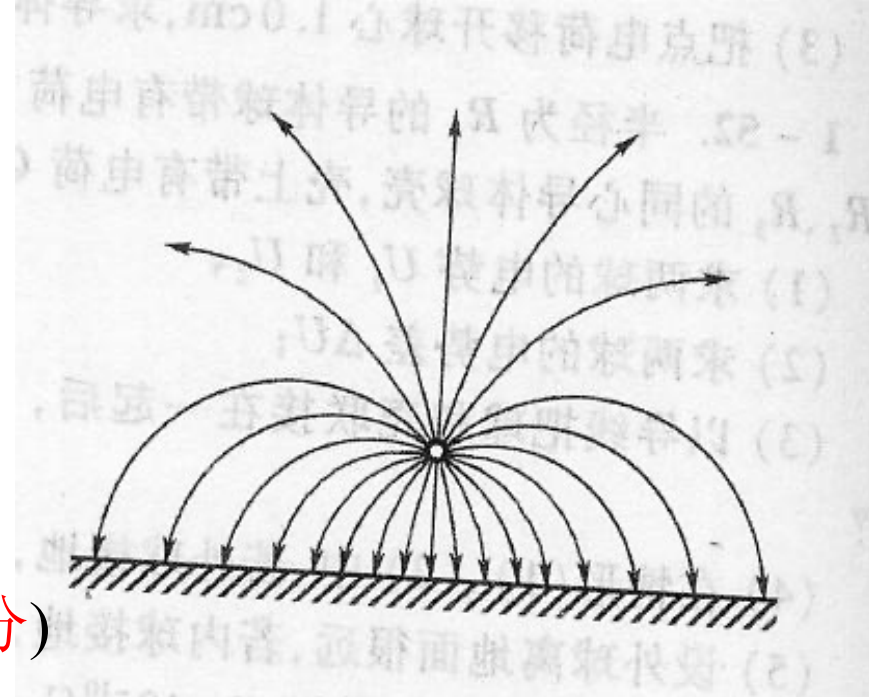
$$U = \frac{\lambda}{2\pi\epsilon_0} \ln \frac{H}{\sqrt{x^2 + (y-H)^2}} - \frac{\lambda}{2\pi\epsilon_0} \ln \frac{H}{\sqrt{x^2 + (y+H)^2}}$$

在地面附近的电场只有Y方向分量(2分)

$$E_y \Big|_{y=0} = - \frac{\partial U}{\partial y} \Big|_{y=0} \overset{\text{2分}}{=} \frac{\lambda}{2\pi\epsilon_0} \frac{y-H}{x^2 + (y-H)^2} \Big|_{y=0} - \frac{\lambda}{2\pi\epsilon_0} \frac{y+H}{x^2 + (y+H)^2} \Big|_{y=0} \overset{\text{2分}}{=} - \frac{\lambda}{\pi\epsilon_0} \frac{H}{x^2 + H^2} \overset{\text{2分}}{}$$

$$\sigma = \epsilon_0 E_y \Big|_{y=0} = - \frac{\lambda}{\pi} \frac{H}{x^2 + H^2} \overset{\text{2分}}{}$$

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电荷分布解法二：

在地面附近任意一点的电场是两条线电荷的电场的叠加(2分)
且只有Y方向分量(2分)

$$E_y \Big|_{y=0} = \frac{\lambda}{2\pi\epsilon_0} \frac{-H}{x^2 + H^2} - \frac{\lambda}{2\pi\epsilon_0} \frac{H}{x^2 + H^2} = -\frac{\lambda}{\pi\epsilon_0} \frac{H}{x^2 + H^2}$$

$$\sigma = \epsilon_0 E_y \Big|_{y=0} = -\frac{\lambda}{\pi} \frac{H}{x^2 + H^2}$$